

Clinical Hypertension (Part 1)

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Thank you to prior scholars!

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Session Objectives

1. Develop a strategy to **screen, diagnose, and stage** hypertension.
2. Demonstrate the role of **risk factors** in hypertension.
3. Explain the **basic pathophysiology** of hypertension.
4. Describe various **forms** of hypertension.
5. Illustrate the **pathologic complications** of hypertension.



Case #1

A 65-year-old **black** male with **history of type 2 diabetes mellitus** and **osteoarthritis** presents to the clinic with a **complaint of headaches**. He currently takes **ibuprofen and metformin**. The patient has a **family history of heart failure and hypertension (HTN)**. He lives a **sedentary** lifestyle and **rarely exercises**. His **job is stressful, and he works long hours**, so his diet consists of mostly **fast food**. He has been **smoking** for 35 years and has two **beers** a night. He says he measures his blood pressure at home, and it is **140/80 mmHg**. He brought his cuff in with him today.



Case #1 continued

HR: 97 bpm

T: 98.2 °F

RR: 14 breaths per minute

BP: 145/92 mmHg (L), 143/91 mmHg (R) (*automatic reading*)

Ht: 6' 0 ft

Wt: 250 lbs

BMI: 35.9 kg/m²



Hypertension Definition

Category	Systolic (mmHg)	Diastolic (mmHg)
Normal	< 120	< 80
Elevated	120 – 129	< 80
Hypertension Stage 1	130 – 139	80 – 89
Hypertension Stage 2	≥ 140	≥ 90

If there is a disparity in category between the systolic and diastolic pressures, the higher value determines the stage.



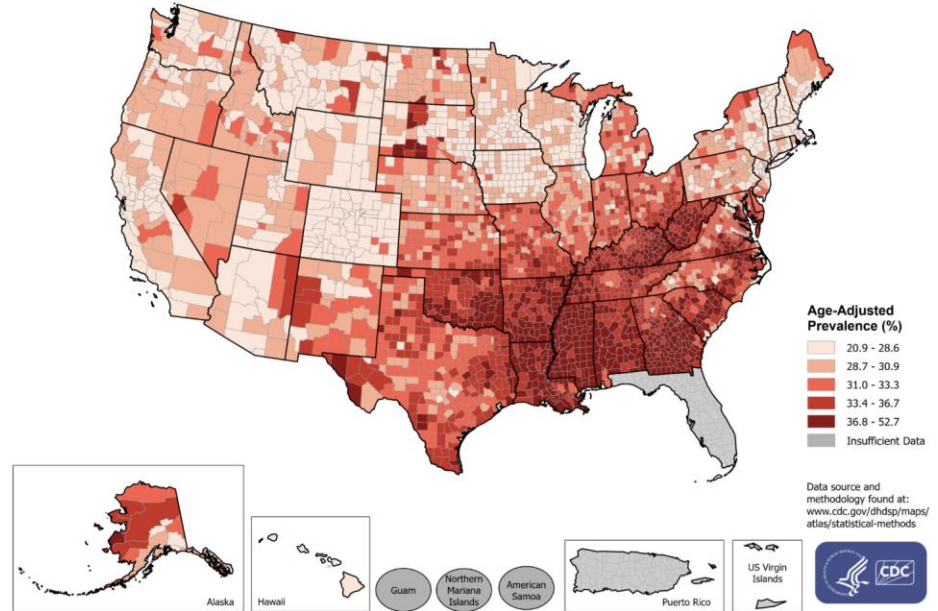
Epidemiology

- HTN is a major risk factor for heart disease and stroke which are leading causes of death in the US
- 49.4% of adults >20 y/o with HTN
- About 1 in 4 adults have controlled HTN
- Hypertension accounts for 1.3 million visits to the emergency department
- “Silent killer”: Deaths per 100,000 US population from essential hypertension and hypertensive renal disease: 13.0

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High Blood Pressure Prevalence (%), 2021
Adults, Ages 18+, by County



How to Diagnose HTN:

Requires:

≥ 2 readings

≥ 2 occasions (in and out of office)

Appropriate technique and cuff size

**Learn to take
and report blood
pressure
accurately**

**Hear nothing
and say 120/80**

**Hear nothing but
report it with obscure
numbers like 121/78
to make it more
plausible and less like
you made it up on the
spot**



How to Measure Blood Pressure

- Office BP measurement (OBPM)
- Home blood pressure measurement
- Ambulatory BP measurement (ABPM)



Clinic Measurements



Home BP Monitoring



**Ambulatory BP
Monitoring**



When to Screen HTN:

2021 USPSTF recommends screening for high BP in **adults 18 years** or older with OBPM.

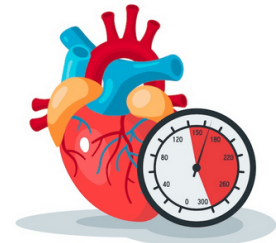
At a minimum:

- Adults with normal blood pressure should undergo yearly BP
- Adults with risk factors (obesity, T2DM) or with prior elevated SBP should be evaluated at least semiannually



Types of Hypertension

- Primary (Essential) HTN
- Secondary HTN
- White Coat HTN
- Masked HTN
- Resistant HTN
- Hypertensive crisis: urgency vs emergency



Types of Hypertension: Primary (Essential) HTN

- “Silent killer” because usually no sx
- 90% of HTN
- Familial; **no clear cause**; interaction between environmental and genetic factors





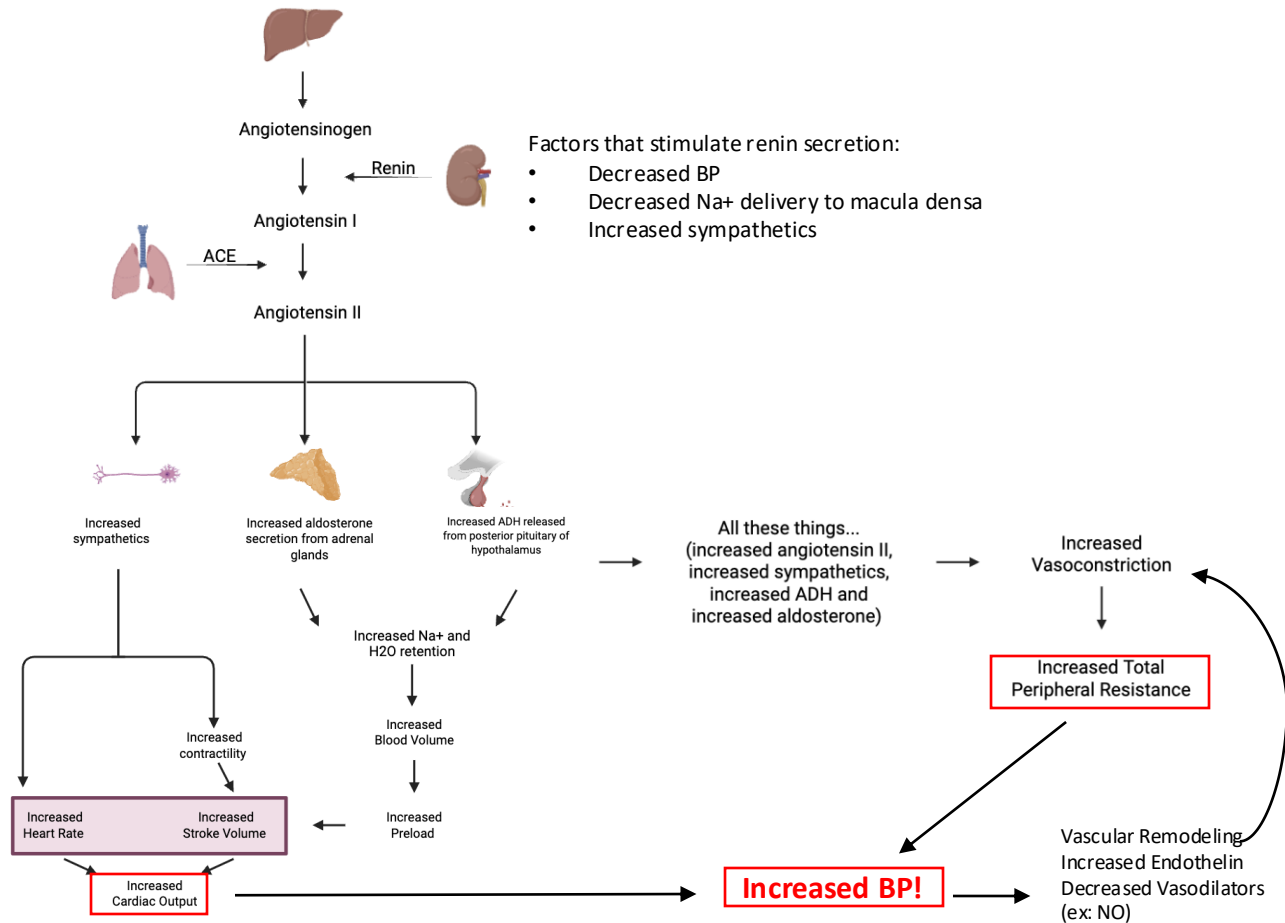
Types of Hypertension: Primary (Essential) HTN

- Pathogenesis:
related to
increased Cardiac Output or
increased Total Peripheral Resistance

$$\text{MAP} = \text{CO} \times \text{TPR}$$

$$\text{CO} = \text{HR} \times \text{SV}$$

Renin-Angiotensin-Aldosterone System



Types of Hypertension: Secondary HTN

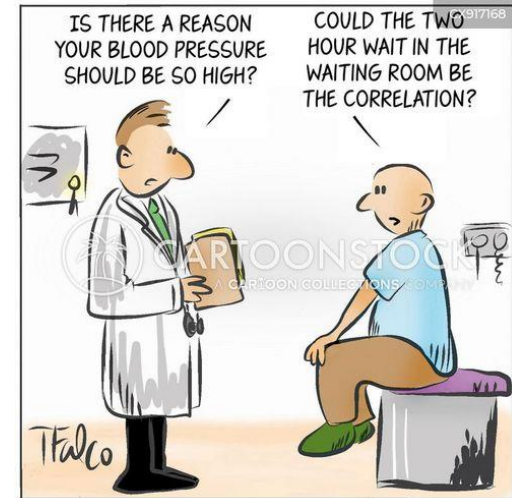
- **Remaining 10% of HTN; known underlying cause**
- **Consider in the following pts:**
 - HTN unresponsive to medications
 - Presents at young age
 - Older pts with new onset HTN
 - Acute worsening of previously well controlled BP
- **Causes:**
 - Cushing's syndrome
 - Hyperparathyroidism, hypothyroidism, hyperthyroidism
 - Aortic coarctation (in young kids, but can be dx in adult)
 - Meds:** Oral Contraceptives, Chronic NSAID use, corticosteroids, decongestants; Illicit Drug Use (ex: cocaine)
 - Pheochromocytoma
 - Primary Aldosteronism
 - Sleep Apnea
 - Stenosis of renal artery and fibromuscular dysplasia



Types of Hypertension:

White coat HTN and Masked HTN

- **White Coat HTN**
 - blood pressure that is consistently **elevated by office readings** but does not meet diagnostic criteria for hypertension based upon out-of-office readings
- **Masked HTN**
 - blood pressure that is consistently **elevated by out-of-office measurements** but does not meet the criteria for hypertension based upon office readings



Indications for ABPM

- Indications: when there is a **discrepancy** between home and office BP measurements
 - Ex: White Coat Hypertension; Masked Hypertension



Types of Hypertension: Resistant HTN

- **Resistant HTN:**
 - remains $\geq 140/90$ mmHg despite optimal use of **3+** antihypertensive medications of different classes, including a diuretic at maximal tolerated dose or when **BP control is achieved through 4+ medications (meaning it is a Controlled Resistant HTN)**
 - **Causes:** non-compliance with medications; secondary HTN; diet



Types of Hypertension: Hypertensive Crisis

- **Hypertensive Urgency:**
 - Severe ($\geq 180/\geq 120$ mm Hg) hypertension without acute end organ damage
- **Hypertensive Emergency**
 - Severe ($\geq 180/\geq 120$ mm Hg) hypertension **with evidence of acute end-organ damage**
 - Encephalopathy, Stroke
 - Retinal hemorrhages, Papilledema
 - MI, HF, Aortic dissection
 - Kidney injury
 - Microangiopathic hemolytic anemia
 - Eclampsia



Types of Hypertension: Hypertensive Crisis

- Causes:
 - Secondary HTN
 - Medication non-compliance
 - Sudden discontinuation of short acting antihypertensive meds (ex: BB, clonidine)
 - MAO inhibitors + tyramine rich foods
 - Recreational drugs (ex: cocaine use)



Primary HTN Risk Factors

- Modifiable:
 - Obesity
 - High-sodium diet/unhealthy diet
 - Excess alcohol intake
 - Physical inactivity
 - Smoking
- Non-Modifiable:
 - Increased age
 - Family hx
 - Race (Black > White > Asian populations)
 - Low Socioeconomic Status



Thinking back to our case #1...

What risk factors did our patient have and which are/are not modifiable?



Thinking back to our case #1...

What risk factors did our patient have and which are/are not modifiable?

NSAID use (modifiable)

Family hx of HTN (not modifiable)

Sedentary lifestyle (modifiable)

Fast food → probably high in sodium (modifiable)

Smoker (modifiable)

Alcohol intake (modifiable)

High BMI (modifiable)

Race (not modifiable)



Q #1:

A 49-year-old woman presents for an annual physical exam. She has no history of hypertension, diabetes, or cardiovascular disease. She exercises regularly and follows a healthy diet. She does not smoke or drink alcohol.

Vital signs are as follows: Blood pressure: 128/84 mmHg; Heart rate: 72/min; Respiratory rate: 12/min; Temperature: 36.9°C (98.4°F)

Repeat blood pressure readings on two subsequent visits are similar. Which of the following is the correct classification of this patient's blood pressure?

- A. Normal
- B. Elevated
- C. Hypertension Stage 1
- D. Hypertension Stage 2
- E. White coat hypertension



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Q #2:

A 60-year-old white male with PMHx of HTN, HLD, T2DM, and heavy tobacco use presents to clinic for a follow up for his high blood pressure. Despite escalating his anti-hypertensive regimen, he has persistently elevated blood pressure readings. His only reported symptom is a headache, and he denies chest pain. He is currently taking 4 antihypertensive medications. His BMI is 25 kg/m². Vitals are unremarkable except his blood pressure is 177/120 mmHg in the right arm and 179/123 mmHg in the left arm. Physical exam is consistent with an upper abdominal bruit. The finding is best explained by which of the following?

- A. Aortic coarctation
- B. Aortic dissection
- C. Abdominal aortic aneurysm
- D. Renal artery stenosis
- E. Obstructive sleep apnea



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Q #3:

A 25-year-old female presents to the emergency department (ED) for an episode of epistaxis associated with extremely elevated blood pressure. Patient has no past medical history. Her blood pressure in the ED is noted to be 170/100 mmHg. Her BMI is 24 kg/m². On physical exam, there are no heart murmurs, abdominal bruits, or thyromegaly. Labs work up, including TSH, serum electrolytes, blood glucose, and creatinine, is normal. Which of the following additional tests would you order next?

- A. Ultrasound of the renal arteries
- B. Polysomnography for sleep apnea
- C. Urine drug screen
- D. Full thyroid panel
- E. Lipid panel



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Evaluation of Hypertensive Patient

- **History: Go through those risk factors!**
 - Determine the presence of precipitating or aggravating factors
 - prescription medications, nonprescription NSAIDs, alcohol consumption
 - Duration of hypertension
 - Previous attempts at treatment
 - Signs of target-organ damage
 - Presence of other known risk factors for cardiovascular disease

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Evaluation of Hypertensive Patient

Physical Exam:

- Many pts are often asymptomatic
- Evaluate for:
 - signs of end-organ damage
 - established cardiovascular disease
 - evidence of potential causes of secondary HTN

General: body fat distribution, alertness

Fundoscopy: flame hemorrhages, arteriovenous nicking, papilledema, cotton wool spots (important fundoscopic examination to evaluate for hypertensive retinopathy!)

Neck: carotid auscultation, neck circumference, thyroid

Heart: size, rhythm, sounds

i.e. **S4 gallop – suggest LVH**; systolic murmur (and HTN proximal and low BP distal) – coarctation

Lungs: rhonchi/wheezes, rales

Abdomen: renal mass (ADPKD), **abdominal bruits (may suggest renal artery stenosis)**, femoral pulses

Extremities: peripheral pulses, edema

Neurologic: visual changes, weakness, confusion



Evaluation of Hypertensive Patient

Lab testing: should be performed in all patients with newly diagnosed hypertension

Electrolytes (including Ca^{+} , Na, K)

BUN/serum creatinine (to calculate the estimated glomerular filtration rate)

Fasting glucose

Fasting lipid profile

HbA1c

Urinalysis

Complete blood count

Thyroid-stimulating hormone

Electrocardiogram

Calculate [10-year atherosclerotic cardiovascular disease risk](#)



Q #4:

A 43-year-old male presents to the office for initial evaluation of high blood pressure. He exercises regularly and follows a low salt diet however his blood pressure continues to be around 154/86 mmHg in office, including today. He has no symptoms or past medical history. His BMI is 26 kg/m². On physical exam, no vascular bruits are heard, peripheral pulses are +2 and symmetrical, and heart and lung exams are unremarkable. ECG shows normal sinus rhythm. Complete blood count and complete metabolic panel are normal. Which additional study would you recommend for the patient?

- A. 24-hour cortisol excretion
- B. Lipid panel and urinalysis
- C. Renal ultrasound
- D. Aldosterone levels
- E. Polysomnography



Q #4:

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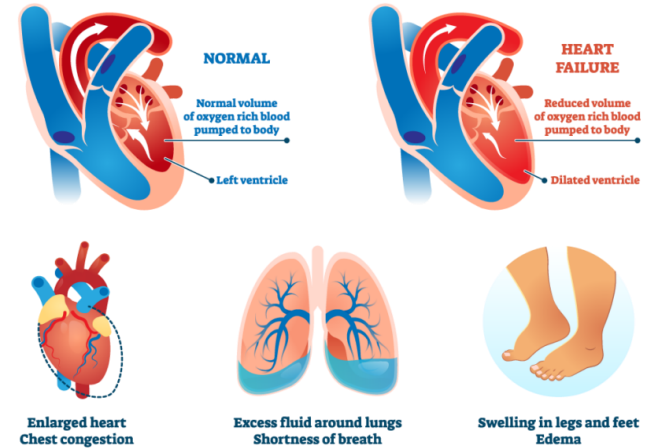
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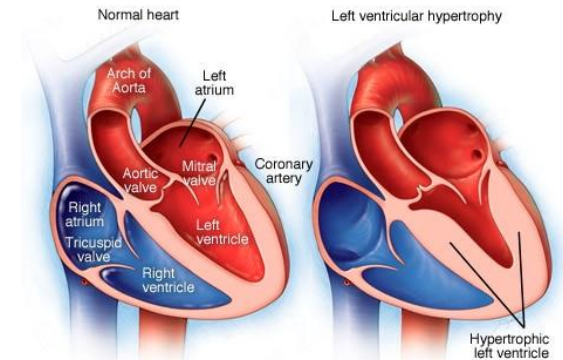
Complications of HTN: Heart

- Left Ventricular Hypertrophy
- Heart failure
- Cardiomyopathy
- Ischemic Heart Disease
 - CAD, MI
- Atrial Fibrillation
- Aortic Aneurysm
- Aortic Dissection
- Carotid Artery Stenosis
- **Managing hypertension is the most prevalent modifiable risk factor for premature cardiovascular disease**

CONGESTIVE HEART FAILURE



<https://sagefoc.com/blog/4-stages-congestive-heart-failure/>

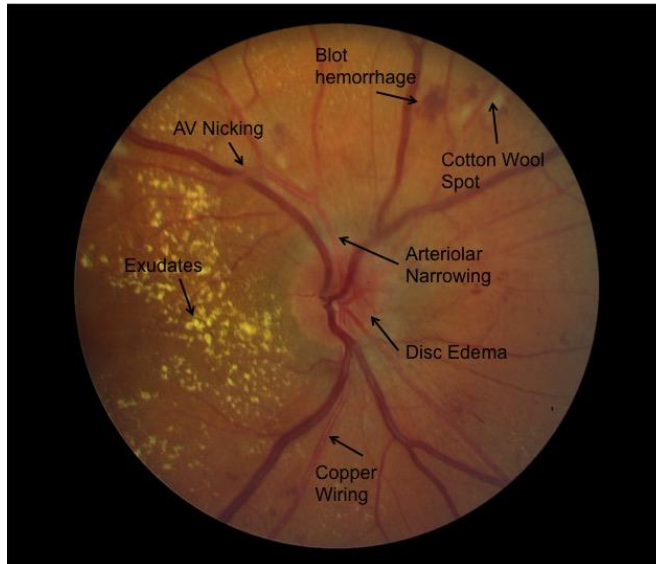


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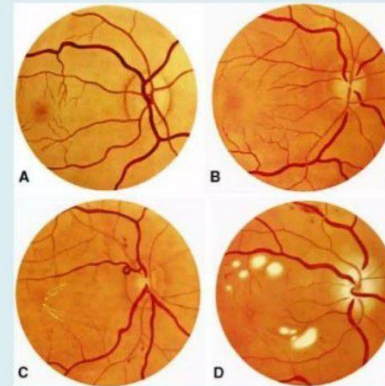
<https://www.mayoclinic.org/diseases-conditions/left-ventricular-hypertrophy/symptoms-causes/syc-20374314>

Complications of HTN

- **Brain:** Stroke, TIA, cognitive impairment, intracerebral hemorrhage
- **Eyes:** Hypertensive retinopathy



Complications: Retina

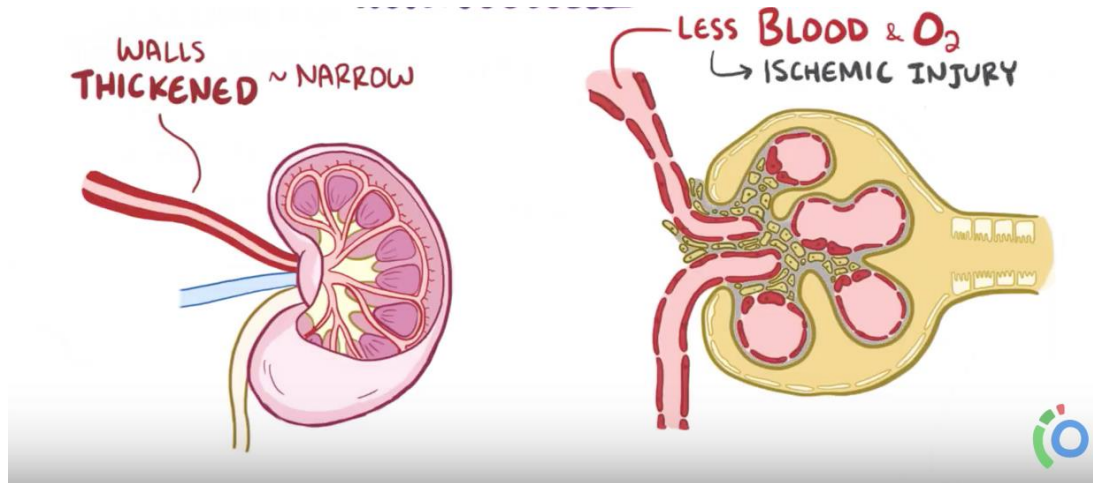


- **Grade 1**
 - Arteriolar thickening
 - Tortuosity
 - Increased reflectiveness - **silver wiring**
- **Grade 2**
 - Grade 1
 - Constriction of veins at arterial crossings - **arteriovenous nipping**
- **Grade 3**
 - Grade 2
 - **Retinal ischaemia** - flame-shaped or blot haemorrhages and 'cotton wool' exudates
- **Grade 4**
 - Grade 3
 - **papilloedema**



Complications of HTN Cont'd

- **Kidneys:** Hypertensive nephrosclerosis

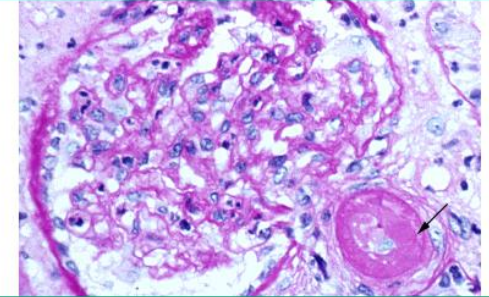


<https://www.youtube.com/watch?v=fv53QZRk4hs>

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Light micrograph showing acute hypertensive nephrosclerosis (formerly "malignant nephrosclerosis")



Light micrograph reveals fibrinoid necrosis in the preglomerular afferent arteriole (arrow) in acute hypertensive nephrosclerosis (formerly "malignant nephrosclerosis"). The normal muscle layer of the media has been replaced by the fibrinoid material.

Courtesy of Helmut Rennke, MD.

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Summary

- Understanding the definitions of hypertension and screening guidelines of a patient will allow for optimal treatment.
- Questions should be asked about the patients modifiable and non-modifiable risk factors as well as modification and drug use to guide the diagnosis of hypertension
- Diagnosing different types of hypertension is important for proper treatment.
- The physical exam should encompass multiple systems as complications of hypertension are systemic and cause devastating effects on health.



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